SERIES IV · FIELD PAPERS

Rule of Life

Penguin Economics, Regenerative Reciprocity, and the Ledger of the Commons

Constitutional Field Paper · Economic Field Paper · Commons Governance Note
Publisher: Spiralweb Papers
Accountable knowledge steward: Lars A. Engberg
v1.3.4 · 18 October 2026 · CC BY 4.0

Editorial practice: AI-assisted and human-responsible. Method and responsibility.

Executive summary

Penguin Economics is not philanthropy. In a donor-led philanthropic relationship, private surplus is the starting point for support serving the public good. Penguin Economics begins with shared exposure and asks how life-supporting conditions can remain viable across soil, water, stewards, communities, institutions, and future generations.

The economic family proposed here is a **Planetary Stewardship Economy**: a polycentric flow architecture in which support enters real places through documented relations, passes through membranes of consent, stream separation, review, non-capture, and field tempo, and may circulate outward only when genuine abundance has emerged without debt, ownership, extraction, or control.

Its basic unit is not the donor, investor, consumer, beneficiary, platform, or project. Its basic unit is the **stewarded living field**: a place and its people, ecological conditions, obligations, evidence, memory, and capacity to continue.

The paper's economic contribution is to connect five disciplines:

- **Penguin Economics**: rotate warmth toward exposure; no steward or place should remain permanently at the cold edge.
- **Regenerative Reciprocity**: support enters to strengthen a field; abundance circulates only after life is stable enough to give.
- **PG Ledger**: field conditions, burdens, evidence, roles, flows, uncertainty, and corrections are recorded and connected without collapsing them into one score.
- **Earth Time**: money, reporting, technology, and institutional ambition adapt to the regenerative tempo of living systems.

- **Rule of Life**: economy, law, governance, and technology remain bound to the conditions that make life possible, while Rule of Law continues to protect against arbitrary power.

The architecture is not governed by the ledger. A ledger cannot govern. It can preserve memory, make burdens visible, support review, and connect evidence to decisions. **Human stewards and legitimate human institutions govern.**

The founding case also asks whether human-held AI inquiry can help people articulate lived experience and act together while they retain authority over their experience and place.

Citizen science observes.
13×13 structures inquiry.
AI assists.
PG Ledger records and connects.
Human stewards govern.
Rule of Life provides the constitutional horizon.

The paper's commons theory moves from Tragedy of the Commons, through Romance of the Commons, toward a Ledger of the Commons. The ledger is not offered as a substitute for devotion, trust, local knowledge, or lawful judgement. It gives shared life enough form to remember burden, resist capture, receive correction, and remain governable without becoming cold bureaucracy.

Romance without ledger can become fog. Ledger without romance can become dead bureaucracy. Together they can become stewardship.

The paper's central thesis is:

> **In a polycrisis, the primary economic task is not to maximise return from living systems, but to restore and govern the conditions that allow life, law, food, health, community, and future economy to continue.**

Reader’s map

This is a constitutional and economic synthesis. It does not claim to replace ecological economics, commons theory, feminist and care economics, post-growth thought, regenerative economics, wellbeing economics, environmental law, or the detailed operational papers within the Spiralweb architecture. It asks one operational question:

> **How can life-supporting stewardship become observable, structured, reviewable, and protected against capture — without surrendering the field to money, data, law, technology, or any single metric?**

PG Ledger is a protocol layer, not merely a database. The economic and relational questions surrounding its possible use are developed in Report 06 — Regenerative Reciprocity. The current epistemic and field-observation method belongs to Knowing From the Ground. The monthly human governance reading belongs to Report 04 — Penguin Dashboard. This paper brings those elements together only where they are needed to state the economic and constitutional proposition.

The account also remains answerable to the human and material conditions in which it was made. Whether the practice helps other stewards is an open question for their own experience, correction, and ability to sustain the work.

The architecture does not require the whole economy, field, or knowledge problem to be solved before work can begin.

Its practical unit is a **bounded next step**: sufficiently grounded to act, sufficiently modest to remain revisable, and sufficiently documented that consequence can become part of the next round of learning.

Section 15 brings this down to a short set of questions about the people, species, and places affected; the evidence and uncertainty; who decides and carries the burden; the scope of the action; and when those affected can question its consequences. This is where the constitutional test meets a living field.

This is not an argument for acting without knowledge. It is a discipline for acting without pretending to possess complete knowledge.

Contents

1. Part I — The Economic Proposition
2. 1. Why philanthropy is not the root category
3. 2. The Planetary Stewardship Economy
4. 3. Regenerative Reciprocity and the entry of money
5. 4. Field first, ledger second, credit third
6. 5. Earth Time and life-supporting value
7. 6. Penguin Economics and the thermodynamics of exposure

8. Part II — Making Stewardship Legible
9. 7. Non-compensatory streams and capture resistance
10. 8. PG Ledger: records, relations, and correction
11. 9. Situated inquiry: citizen science and 13×13
12. 10. Moral Biology and institutional carrying capacity
13. 11. Penguin Dashboard and accountable assistance
14. Part III — The Constitutional and Commons Horizon
15. 12. The Golden Rule under ecological conditions
16. 13. Neighbouring as bioregional practice
17. 14. Rule of Law and Rule of Life
18. 15. Rule of Life is not ecological exception
19. 16. Tragedy, Romance, and Ledger of the Commons
20. 17. Federation, failure modes, finance, and structural decency
21. Closing diagnostic
22. References

PART I

The Economic Proposition

1. Why philanthropy is not the root category

Philanthropy is commonly understood as the voluntary promotion of human welfare through private initiatives for public good. It can relieve hunger, fund schools, protect forests, support research, sustain culture, and keep fragile initiatives alive. The argument here is not that philanthropy is false or morally empty.

The distinction is architectural.

A donor-led philanthropic relationship often begins with private surplus. One party has accumulated capital, institutional reach, discretionary freedom, reputation, or technical capacity. Another is identified as needing support. The philanthropic gesture moves surplus from the one who has to the one who needs. The important question is how much authority over purpose, conditions, and continuation remains with the giver. Trust-based philanthropy already addresses that imbalance through practices such as multi-year unrestricted funding and acting on feedback. Penguin Economics joins this concern while asking how support can begin from shared exposure and the capacity of a living field.

Penguin Economics begins elsewhere. It begins with **shared exposure**. It asks where the cold edge is; who or what is carrying too much; which soils, waters, communities, institutions, and stewards are losing regenerative capacity; and how warmth, time, attention, safety, knowledge, and money must rotate before the system breaks.

> **Philanthropy asks: who gives to whom?**
>
> **Penguin Economics asks: where is life exposed, and how must capacity rotate toward that exposure before collapse?**

This is why philanthropy may be an input into the proposed economy without becoming its root grammar. Once philanthropic money enters a stewarded field, it must pass through the same membranes as every other form of support: consent, purpose clarity, stream separation, ledger visibility, field tempo, burden awareness, review, correction, and protection against ownership or narrative capture.

The grammar is stewardship.

This change of grammar also changes the basic economic subject. The recipient is not a passive beneficiary. A local steward, community, school, farm, wetland group, municipality, or river association may carry knowledge, responsibility, practical intelligence, intergenerational memory, and ecological work that the funder does not possess. Support is not a transfer from competence to lack. It enters a relation in which different capacities meet and in which the field itself has standing.

The test is therefore not the purity of the giver's intention. The test is what the support does to the field:

- Does it strengthen ecological and human carrying capacity?
- Does it increase or reduce local agency?
- Does it make hidden labour visible?
- Does it respect the field's absorptive tempo?
- Can those affected contest, pause, redirect, or refuse it?
- Does it leave the field more able to remain itself?

A gift that overwhelms, controls, accelerates, extracts story or data, centralises judgement, or makes local practice perform for distant expectations may be generous in intention and still become capture in effect.

2. The Planetary Stewardship Economy

The umbrella term proposed here is **Planetary Stewardship Economy** — in Danish, planetarisk forvalterøkonomi.

The term changes the central actor. The investor, consumer, donor, entrepreneur, beneficiary, state administrator, and platform do not disappear. But none is sovereign. The central figure is the steward: the person, group, field, or local institution carrying responsibility for the continued viability of life-supporting conditions.

The primary economic act is not buying, selling, donating, owning, investing, or extracting. It is **carrying life over time**.

When a farmer changes practice so soil structure improves, water is held, biodiversity returns, food resilience grows, family viability strengthens, and local learning expands, this is not merely a private agricultural decision. It is governance at plot scale. A ten-square-metre observation, restored wetland edge, chinampa, food forest, seed-saving practice, composting cycle, neighbourhood repair, or school garden may become planetary when it contributes to a shared, reviewable capacity to protect life-supporting systems.

The size of an intervention does not determine the scale of its meaning. Planetary does not mean centralised or enormous. It means situated action understood inside shared ecological dependency.

“Planetary” therefore does not mean that one economy, institution, protocol, knowledge system, or centre should govern the planet.

It names a condition rather than a jurisdiction: situated economies already participate in ecological, material, climatic, technological, and social relations whose consequences travel beyond the places in which they originate.

The planetary condition is shared.

Authority remains situated and polycentric.

The proposed economy is:

- **Regenerative**, because its test is whether practice supports soil, water, food, biodiversity, human viability, trust, learning, and future capacity.
- **Stewardship-based**, because the core actor is the one carrying responsibility in place.
- **Polycentric**, because legitimate authority is distributed across people, local circles, institutions, public law, review bodies, and connected places rather than placed in one centre.
- **Ledger-supported**, because evidence, roles, burdens, uncertainty, flows, thresholds, decisions, and corrections must remain visible enough for review.
- **Non-extractive**, because a field is not required to produce outward value before it can do so without depletion.
- **Flow-based**, because its aim is calibrated circulation rather than permanent accumulation.
- **Non-compensatory**, because apparent success in one stream cannot erase collapse in another.

The architecture is not anti-money. It is anti-capture. It is not anti-capital. It is against capital becoming the sovereign grammar of the field. It is not anti-market in a simplistic sense. It refuses to let price become the final judge of life.

It stands in conversation with ecological economics, steady-state and post-growth economics, degrowth, doughnut economics, wellbeing and beyond-GDP approaches, feminist and care economics, commons theory, regenerative economics, solidarity economy, circular economy, community wealth building, and foundational economy. These traditions have already demonstrated that GDP, price, profit, and growth do not provide a

complete account of value; that care and social reproduction are economic; that the economy is nested within the biosphere; and that commons can be governed through rules, monitoring, conflict resolution, and polycentric institutions.

The distinct contribution offered here is not to supersede those traditions. It is a field-operational proposition:

> **Stewardship must become observable, structured, reviewable, and protected against capture — without being reduced to a single metric or transferred away from human judgement.**

This produces a set of core axioms:

1. Life is the primary balance sheet.
2. Money may enter, but it must not become sovereign.
3. No single metric may become sovereign over the field.
4. The field sets the tempo.
5. Steward viability is not external to ecological viability.
6. Legibility requires governance, not merely data.
7. Abundance may circulate only when life is stable enough to give.
8. Rule of Life must remain bound to Rule of Law.
9. Federation, not aggregation, is the scaling logic.
10. AI may assist legibility, but it must not replace responsibility.
11. Ethics is not only what we believe. It is what we can carry.

3. Regenerative Reciprocity and the entry of money

The architecture is not adequately described as market, gift, philanthropy, state redistribution, impact investment, or mutual aid alone. Each may contribute something, but each leaves part of the proposed relation unnamed.

Market form can reduce too much to price. Pure gift can hide obligation, hierarchy, exhaustion, or dependency. Donor-led philanthropy preserves the giver’s discretion over surplus. State redistribution depends on public authority and legal mandate. Impact investment commonly attaches ownership, return, or claimable impact to capital. Mutual aid may be deeply relational but does not by itself provide durable evidence, accounting, conflict handling, institutional memory, or safeguards against hidden burden.

The operational flow principle is **Regenerative Reciprocity**.

Support enters a living field in order to restore or strengthen its capacity. Only when genuine abundance appears may capacity circulate outward. It does not circulate as repayment of debt, investor return, ownership yield, moral obligation, or compulsory gratitude. It circulates as life-bearing capacity released from one strengthened field toward another exposed field.

> **Relation is primary. Money is accountable. Stewardship is documented. Abundance circulates only after life is stable enough to give.**

Report 06 — Regenerative Reciprocity develops a public working inquiry around AnchorPoints, entry conditions, three-stream distinctions, possible outward circulation, review, and institutional safeguards. It does not establish an operating standard. The purpose here is to state the economic role of that inquiry.

Money does not arrive pure

The first practical objection is unavoidable: if this is not conventional market logic, ordinary philanthropy, state redistribution, or impact investment, where does the money come from?

In early phases, most money will come through existing economic channels: households, memberships, donations, foundations, companies, public grants, institutional partnerships, fees, advisory work, licensing, banks, markets, ordinary fiat currency, and the infrastructures of the present economy.

This is not a contradiction. It is the condition of transition.

The question is not whether money is pure before entry. The question is whether it can enter a living field without becoming sovereign over it. Regenerative Reciprocity does not purify money by intention. It disciplines money through architecture.

Money must pass through membranes:

- clear purpose and source;
- separation of ecological, steward, and governance realities;
- no ownership claim over land, data, story, future production, or local decision;
- no hidden control or conditionality;
- no acceleration beyond absorption capacity;
- no substitution of donor narrative for field truth;

- no reporting burden that overwhelms the work;
- correction, contestability, and stop rights;
- no ecological ambition financed by human depletion.

The first money may come from the old economy. The discipline is that it must not reproduce the old economy inside the field.

If money enters without membranes, it becomes capture. If it enters and demands speed beyond the field's capacity, it becomes capture. If it buys control over governance, knowledge, land, story, or future value, it becomes capture. If local stewards must perform the funder's language rather than report field truth, it becomes capture.

PG Ledger supports capture resistance by connecting money to purpose, evidence, responsibility, burden, consent, decision, and correction. It cannot make money innocent, and it cannot govern the relation by itself. Human beings and institutions must judge whether the membranes are holding.

The transition sequence

1. Money enters from existing sources.
2. Its purpose and conditions become visible.
3. Human stewards and legitimate institutions apply consent, stream separation, non-capture, review, and field-tempo discipline.
4. The field uses support to strengthen land and ecology, steward viability, and governance capacity.
5. Only when real abundance exists may capacity move outward through Regenerative Reciprocity.

This is a transition architecture for stewardship capacity. Its aim is not merely to finance cleaner assets or improved sustainability performance inside existing balance-sheet logic. Its aim is to strengthen the local and institutional capacity to carry life-supporting systems over time.

4. Field first, ledger second, credit third

Carbon accounting, biodiversity accounting, MRV, insetting, nature-related disclosure, contribution claims, sustainability reporting, transition finance, and tradable credits are not identical. They belong to different legal, voluntary, financial, and organisational regimes. They should not be collapsed into one category.

They do, however, share a threshold: they depend on trustworthy field legibility.

A market cannot internalise what it cannot see. Nor can it responsibly internalise a living field if it sees only one metric.

Carbon may be a valid dimension of change, but carbon is not life. A project may improve carbon balance while degrading biodiversity, exhausting stewards, reducing food sovereignty, centralising land control, or reorganising local practice around external methodologies and buyer claims.

The problem is not carbon accounting as such. The problem is **carbon sovereignty**: the elevation of one metric into the master language of the field.

PG Ledger can hold carbon, biodiversity, financial, social, and institutional evidence inside a wider account of life-supporting conditions. It does not make tradable conversion the default purpose of observation.

> **Field first.**
> **Ledger second.**
> **Credit third — if at all.**

Carbon and biodiversity evidence may enter the ledger. Tradable credits do not follow automatically. A credit, claim, or financial instrument should never define in advance what the field is for, what must be observed, or which burdens may disappear.

The ordering protects four distinctions:

- observation before claim;
- field purpose before buyer demand;
- human and ecological viability before conversion;
- the right to refuse before the pressure to monetise.

A credit system is not a passive receiver of better evidence. Platforms, buyers, auditors, standard-setters, intermediaries, and financiers carry incentives toward volume, comparability, standardisation, claimable value, and tradable units. Those incentives may improve discipline in some contexts; they may also make the field more legible to capital than to the people who live there.

The correct response to a distorted conversion pressure may be refusal, pause, redesign, or withdrawal. If credit demand would weaken field truth, local sovereignty, steward viability, food security, biodiversity integrity, governance capacity, or Earth Time, better packaging is not the answer.

Credits should arise from stewardship only where the relation remains legitimate. Stewardship must never be reorganised around credits.

5. Earth Time and life-supporting value

Many institutions act as if living systems can be accelerated by command.

A tree does not grow faster because a spreadsheet requires quarterly performance. Soil does not rebuild according to an investor exit horizon. Trust cannot be compressed without distortion. A wetland does not regenerate within a political term because a strategy document demands visible results.

The dominant economy operates through financial time, electoral time, project time, credit time, platform time, media time, and reporting time. Living systems operate through soil time, water time, forest time, healing time, trust time, steward time, and generational time.

The mismatch is one of the deepest forms of extraction.

> **ROI time asks when value returns.**
>
> **Earth Time asks when life is ready.**

Earth Time is not romantic slowness. It is temporal governance. It requires pause gates, absorption checks, staged support, locally appropriate review intervals, and the right to slow down when money, reporting, technology, institutional ambition, or public visibility begin to outrun field metabolism.

Precise patience is not passivity. It distinguishes between readiness and pressure, rest and stagnation, learning and delay, absorption and flood.

Institutional capital may carry pension, insurance, endowment, sovereign, or intergenerational obligations. Such institutions can sometimes think in long horizons, yet those obligations often remain held inside short-term valuation systems: benchmarks, liquidity requirements, annual reporting, market pricing, narrow fiduciary interpretations, and political pressure compress the future into the present.

A Planetary Stewardship Economy invites capital into a different time discipline: not rapid extraction from living systems, but patient alignment with the time required for life-supporting capacity to regenerate.

Value in a polycrisis

In a polycrisis, value changes meaning. What appears unprofitable from a narrow return perspective may restore the conditions on which all future return depends.

A field that rebuilds soil, retains water, strengthens biodiversity, feeds people, reduces disaster exposure, supports local stewardship, and increases trust may not generate fast financial return. It may nevertheless prevent loss, restore optionality, reduce dependency, and protect the living balance sheet beneath formal economy.

Profit is not false. It is incomplete.

Profit may indicate viability only where soil, water, people, institutions, and future capacity are not being depleted to produce it.

Life-supporting value is value that protects, restores, or increases the conditions on which future economic, social, ecological, legal, and democratic life depends. It often appears first as avoided loss, restored capacity, resilience, or an option kept open rather than as immediate revenue.

If an accounting system cannot see those forms of value, the accounting system is poor. The field is not.

6. Penguin Economics and the thermodynamics of exposure

Penguin Economics is not an economics of kindness. It is an economics of exposure, warmth, rotation, and viability.

The thermodynamic language is analogical, not a formal physical model. It makes depletion, safety, overload, capacity, and rotation visible as economic realities without reducing ecological or social life to physics.

Penguin huddling offers an image of shared warmth under exposure. The human norm drawn from it is that relief and responsibility should circulate, so the same people do not remain at the exposed edge. The image does not establish a biological rule of equal rotation.

In this language, **cold** may mean scarcity, threat, debt pressure, ecological fragility, isolation, institutional abandonment, unsafe work, or human exhaustion. **Warmth** may mean food, water, safety, time, income, attention, trust, learning, relational protection, and the ability to continue.

> **Where is the cold edge, and how must warmth rotate so the living system does not break?**

This is not altruism organised around a heroic giver. It is distributed survival intelligence.

- No steward should be permanently assigned to the cold edge.
- No ecological ambition should be financed by human depletion.
- No commons should be protected by burning out those who love it.
- No institution should call a field successful while the people carrying it are collapsing.

Steward viability is not an administrative welfare concern added after ecological work. It is part of ecological work. A burnt-out steward is not an externality. Burnout is field data.

The same logic applies across scale. A wealthy region may externalise cold into supply chains. A digital platform may concentrate warmth while distributing precarity. A climate programme may move ecological ambition toward a place while leaving administrative overload with local actors. An institution may appear green by making its burden invisible elsewhere.

Penguin Economics asks where warmth accumulates, where exposure is assigned, and whether the system contains a real capacity to rotate care, money, responsibility, rest, and decision-making power.

PART II

Making Stewardship Legible

7. Non-compensatory streams and capture resistance

The operational architecture keeps three condition readings visible. These describe what is happening, not what has been spent. Resource destinations A (Land and Ecology), B (Steward Viability), and C (Coordination and Governance) are recorded separately, as explained in GP22 §11. An expenditure in one category does not establish a healthy condition.

1. **Land / Life**
2. **Steward Viability**
3. **Governance / Commons**

Condition reading	Typical evidence	Warning signals	Human response
Land / Life	soil, water, biodiversity, food capacity, habitat, regeneration	erosion, contamination, water stress, biodiversity loss, declining resilience	pause, repair, field redesign, technical and local review
Steward Viability	time, income, rest, safety, health, agency, succession, relational capacity	burnout, unpaid overload, unsafe work, hidden dependency, loss of voice	reduce burden, redirect funds, rotate responsibility, slow tempo, provide support
Governance / Commons	roles, consent, evidence quality, conflict handling, review, reporting capacity	unclear authority, capture risk, weak records, conflict escalation, reporting overload	clarify responsibility, strengthen review, limit inflow, invoke correction or pause

These are not merely three budget lines. They are three different realities that must remain visible.

A field may improve ecologically while the steward becomes exhausted. That is not success. A steward may be paid while the land remains degraded. That is not success. A governance system may produce excellent reports while water, soil, people, and trust remain under pressure. That is not success.

The streams are **non-compensatory**. Strength in one cannot morally or analytically cancel collapse in another.

A biodiversity measurement can be true about what it actually measures without being a complete reading of ecological viability. Species richness, the abundance of native populations, genetic diversity, ecosystem extent, connectivity, and risk of collapse answer different questions. A count may rise while invasive species spread or native populations decline. Its meaning also depends on the area, season, method, and reference condition chosen; a recent baseline can conceal earlier loss. The response is to use sound ecological methods and relevant expertise, state what each finding can support, and retain binding ecological limits where they apply. A good result in another stream cannot cancel credible evidence of ecological harm. The Convention on Biological Diversity's monitoring framework likewise tracks distinct components rather than relying on one biodiversity score.

Conventional financial accounting consolidates monetary positions effectively, but ecological depletion, human overload, governance fragility, and loss of future capacity often remain outside its primary statements or appear as risks that may be compensated elsewhere. The three-stream discipline refuses that disappearance.

> A green number cannot cover a red body.
> A strong yield cannot cover collapsing water.
> A good story cannot cover missing evidence.
> A payment cannot buy truth.

Non-compensation is an accounting discipline, not an automatic decision rule. Real choices arise under pressure. A steward may have urgent viability needs while ecological repair also matters. Governance capacity may need strengthening before further money can responsibly enter. A land intervention may need to pause because the human or institutional membrane is too thin to carry it.

Situated human judgement remains necessary. The discipline requires people to name the conflict, document what is prioritised and deferred, identify who carries the burden, and return to the unresolved stream rather than declaring it solved.

The principle is not that all streams can always be equally satisfied at once. The principle is that no stream may disappear from view.

8. PG Ledger: records, relations, and correction

PG Ledger is the protocol and memory layer that connects situated evidence, interpretation, responsibility, support, decision, and later review.

It is not an autonomous governor, an oracle, a centralised database of truth, or a substitute for law, expertise, local judgement, or democratic authority.

A ledger becomes useful when it can answer:

- What was observed, by whom, when, and under what conditions?
- What is interpretation, and what is direct observation?
- Where is uncertainty, disagreement, or missing evidence?
- Who is affected and who has standing to contest the account?
- What money or support entered, for what purpose, and with what conditions?
- Who made the consequential decision?
- What burden was accepted, transferred, or left unresolved?
- Can the decision be corrected, paused, or reversed?

Legibility is therefore not synonymous with visibility to a platform, funder, government, or market. The question is legibility for whom, for what purpose, under whose governance, and with what right of correction?

A field made highly legible to capital but opaque to local stewards is not well governed. A field made legible through invasive data extraction may lose the very sovereignty the ledger was intended to protect.

PG Ledger provides a surface on which field memory can persist without becoming unquestionable. It should preserve evidence and rationale while keeping correction live. The formal requirements for provisionality, correction rights, stop rights, and named human accountability are developed in The Correction Loop.

The governance sequence is explicit:

> Evidence enters the ledger.
> People interpret it.
> A named human or legitimate human body decides.
> Affected people retain routes to challenge and correction.
> The record remains available for later review.

The ledger supports governance. It does not become the governor.

9. Situated inquiry: citizen science and 13×13

The Ledger of the Commons depends on observations from real places. Citizen science provides an important base, but observation is not automatically truth. Testimony, measurement, images, stories, bodily response, local memory, institutional records, and technical data each carry different strengths and limitations.

The current 13×13 method is developed fully in Knowing From the Ground.

It is not a taxonomy of reality and does not divide existence into thirteen final layers.

It is a **bounded and revisable inquiry grammar** that has appeared in several related forms through the wider work.

One reflective 13×13 approaches a field through inquiry lenses and dimensions. The Gaia GoldBloom 13×13 approaches from situated domains and prompts through which a person or place may begin to notice what matters.

The forms are **homologous rather than identical**.

They preserve recurring inquiry-functions — relation, embodiment, consequence, time, power, knowledge, uncertainty, and correction — while their actual categories may differ according to purpose, place, and period.

Neither is a universal checklist.

No participant is expected to complete 169 cells.

No particular thirteen-fold form is claimed as the final structure of reality.

> **The grammar belongs to the inquiry. Reality exceeds it.**

In practice, a field may begin with one bounded observation — a piece of ground, a water note, a missing pollinator, a steward’s fatigue, conflict in a circle, an image of erosion, a child’s observation, or uncertainty about an AI classification — and deepen inquiry only where relevance, pressure, consent, and capacity justify it.

This is one practical meaning of **pixel precision**.

Precision does not mean making the whole field maximally machine-legible.

It means making an observation or claim sufficiently situated to know where it comes from and what it can legitimately carry:

- what was observed;

- where and when;
- by whom or through which method;
- under what conditions;
- with what uncertainty;
- at what scale;
- and with what standing.

A pixel is therefore a bounded point of contact, not a miniature model of the whole.

The method protects against two opposite errors:

1. **Narrow metric capture**, in which only easily quantified indicators are treated as real.
2. **Unstructured meaning**, in which everything feels significant but nothing becomes contestable or reviewable.

Between them lies a more plural discipline.

Situated observation may already be knowledge without being exhaustive knowledge.

Specialist knowledge may reveal mechanisms unavailable to direct observation without thereby acquiring monopoly over the field.

A hydrologist, ecologist, farmer, child, institution, dataset, historical record, local steward, or AI system may each contribute forms of access whose evidentiary standing differs.

The task is not to make them equivalent.

Nor does the method claim to solve the deeper problem of incommensurability among disciplines, traditions, experiences, and modes of knowing.

Its more practical ambition is to create enough shared form that differences can meet without first being erased.

Situated evidence therefore enters the ledger with source, status, uncertainty, and context. It may be tested against repetition, material consequence, measurement, specialist knowledge, other situated observations, disagreement, or later correction.

A small observation can contribute to shared learning without being inflated into proof or dissolved into centralised expert control.

> **You do not have to know the whole field to enter it responsibly.**

The recurrence of thirteen across parts of the wider architecture should not be mistaken for a claim that reality possesses a privileged thirteen-fold structure.

Through the development of the work, thirteen has repeatedly served a practical design intuition:

> **large enough to hold meaningful plurality; small enough to remain cognitively and relationally inhabitable.**

That intuition remains provisional.

Different 13×13 forms use different axes. Different thirteenth positions may do different work. A Circle of 13 concerns human relational scale and should not be confused with a thirteen-domain observation grammar.

Neither thirteen people nor thirteen categories are universal requirements.

Where another number, scale, set of questions, or existing local form works better, the place should not be forced into the inherited geometry.

The 13×13 is a scaffold for inquiry, not an ontology, credential, score, or condition of participation.

10. Moral Biology and institutional carrying capacity

The economic argument rests on a biological and institutional foundation: ethical action depends on capacity.

People may know what ought to be done and still be unable to do it. Institutions may affirm care, rights, participation, and ecological responsibility while producing harm through overload, fragmentation, narrowed attention, insecurity, and procedural drift.

> **Ethics is not only what we believe. It is what we can carry.**

This matters economically because extraction can appear as strain on regulation and judgement. Pressure can narrow attention, make care more reactive, and make participation more symbolic when institutions are overloaded. Consent may become mere

compliance when stewards are pushed beyond their capacity. When urgency becomes the permanent baseline, opportunities for repair diminish. These connections need examination in the particular field.

A Planetary Stewardship Economy must therefore ask not only how money moves but whether bodies, relationships, institutions, and living systems remain within workable ranges.

This is not softness. It is precision.

A system that depends on exhausted people to demonstrate ecological success is already spending the future it claims to protect. A system that treats unlimited administrative capacity as a free resource hides extraction inside governance.

Stewardship time may be purchased through other paid work, savings, debt, precarious income, or unpaid care. These are real conditions of the work, not private shortcomings to be hidden behind a public story of dedication. The proposed economy must ask who can afford to begin, who remains at the cold edge, what a fair livelihood requires, and whether responsibility can be shared before it becomes another form of extraction. Rest and relief may then become possible; they are not prerequisites a strained steward must demonstrate to deserve standing.

The broader conceptual foundation belongs to Green Paper 01 — Moral Biology and the related Civic Nervous System papers. In this synthesis, its decisive consequence is simple: no Rule of Life can be carried by institutions that continuously destroy the capacity for judgement, consent, care, and correction.

11. Penguin Dashboard and accountable assistance

Penguin Dashboard is a light orientation layer: a recurring human reading of Land / Life, Steward Viability, and Governance / Commons. It helps a circle see whether conditions remain workable, are becoming strained, or require protection and pause.

The Dashboard is not governance by colour. Green, Yellow, Red, and Unknown are not rankings, automated verdicts, or moral scores. They are prompts for attention and response. Unknown must not be translated into Green. A strained or exhausted person retains standing to describe the field and challenge its account; Green is not a condition of being heard. The full instrument is specified in Report 04 — Penguin Dashboard.

Planetary boundaries provide a wider context. The framework identifies nine Earth-system processes. The 2023 update found six beyond the safe operating space; the 2025 Planetary Health Check assessed ocean acidification as the seventh transgressed

boundary. Local fields are nested within these planetary conditions, but the nine boundaries should not become a universal dashboard mechanically imposed on every place.

The discipline is:

> Planetary boundaries provide context and constraint.
> Local stewardship provides the ground.
> PG Ledger preserves evidence and responsibility.
> Penguin Dashboard supports orientation.
> Human stewards judge and act.
> Rule of Life provides the horizon.

AI may assist this work through translation, comparison, pattern recognition, memory support, accessibility, evidence organisation, detection of missing information, and preparation of summaries. It does not verify truth by itself, determine significance, assign Dashboard colours, approve funding, govern a field, or carry legal and moral responsibility.

The purpose of AI is not to remove humans from stewardship. It is to increase the field's capacity to notice, remember, translate, compare, coordinate, correct, and care while keeping responsibility human, situated, attributable, and reviewable.

The positive possibility is equally important.

AI can sometimes hold more relations in view than any one participant can actively carry at once: comparing observations across time, translating between technical and ordinary language, recovering earlier context and corrections, locating relevant specialist knowledge, and noticing possible patterns across distant but related fields.

Used well, this can enlarge the shared field of inquiry and make previously difficult bridges easier to build.

One possible test of this collaboration is its return to the person and place from which an account began. Can someone recognise what fits, name what does not, describe the burden in their own words, and change the next step? A clear AI synthesis is provisional until people can question it against bodily experience, relationships, material conditions, other knowledge, and later consequences. Its precision is useful when it helps that return; it becomes dangerous when its fluent distinctions close the inquiry.

But synthesis must not become sovereignty.

A pattern detected across many places does not overrule a place. A representation of a bioregion does not contain the bioregion. A model of the whole does not acquire authority over the people and living systems from which its representations were derived.

AI may expand legibility and connection while remaining one participant in a larger field of knowing.

> **AI may expand the inquiry. It does not own the field.**

The wider political economy also matters. If AI increases productivity while displacing workers, degrading professional judgement, concentrating wealth, or weakening local institutions, Penguin Economics asks where the cold edge moves. Work is not only income. It is dignity, learning, identity, contribution, relationship, and civic participation.

Stewardship must not become a romantic substitute for wages, rights, rest, professional dignity, social protection, or secure livelihoods. The point is not to move people from paid work into unpaid meaning. The point is to recognise that the largest undone work of the century includes the care and regeneration of habitat, and to ask whether an economy will support people to give to life rather than treating their participation as cost or surplus labour.

The test is not whether AI can perform a task. The test is whether its use strengthens or weakens the life-supporting capacity of the field.

PART III

The Constitutional and Commons Horizon

12. The Golden Rule under ecological conditions

The Golden Rule is usually expressed as a moral relation between persons: treat others as you would wish to be treated.

Penguin Economics extends this relation into ecological and economic reality. The “other” is not only another person. The relation includes soil, water, seeds, wetlands, pollinators, children, future generations, local stewards, and exhausted fields — not because non-human life is simply the same as a human legal subject, but because the conditions of life are constitutive of every human relation.

A planetary formulation is:

> **Do not impose on soil, water, bodies, communities, or future generations a burden you would not accept as the condition of your own life.**

This is structural realism, not sentimental morality. Humans are inside the systems they affect. To degrade water is to degrade the future body. To exhaust soil is to exhaust the future table. To burn out stewards is to burn out governance. To treat the commons as an external resource is to misunderstand dependency.

The Golden Rule becomes a ledger discipline without replacing rights, procedure, expertise, or lawful review:

> **No field may be asked to carry a burden that the system refuses to see.**

If apparent gain in one column produces hidden burden in soil, water, body, community, institution, or future viability, the account remains incomplete.

13. Neighbouring as bioregional practice

Rule of Life becomes practical first as neighbouring.

A neighbour is not only the person next door. Neighbouring includes the upstream farmer, downstream community, shared aquifer, migratory species, food system, waste pathway, school, municipality, and future resident whose conditions are shaped by present action.

Bioregional stewardship makes dependency concrete. Water, food, repair, weather, grief, conflict, and care connect people before abstract identity divides them.

This does not abolish conflict. Land, water, labour, access, memory, authority, and competing needs remain contested. Stewardship changes the medium in which conflict is held. It can be documented, slowed, mediated, corrected, and returned to shared conditions.

A bioregion is also an **epistemic meeting ground**.

No single resident, discipline, institution, dataset, or model can know a watershed, food system, coastline, soil ecology, city-region, or migratory corridor in full.

Hydrology, geology, ecology, farming practice, public administration, historical memory, local observation, cultural meaning, remote sensing, and other forms of knowledge may each reveal different aspects of the same living field.

They are not automatically commensurable, and the purpose is not to collapse them into one universal language.

Bioregional stewardship requires something more modest: enough shared ground for different forms of access to meet, remain distinguishable, challenge one another, and become mutually corrigible where connection is possible.

The bioregion is therefore not merely another administrative container. It is a living field of relations that repeatedly crosses the categories through which institutions ordinarily divide knowledge and responsibility.

A locally favourable reading cannot cancel damage in a connected watershed or the exhaustion of those maintaining the practice. A local Green does not compensate for a wider-field Red. Where the larger field is Unknown, the uncertainty remains visible while the people involved decide what the next bounded inquiry can carry.

The Golden Rule under ecological conditions becomes practical: do not impose on a neighbour, watershed, soil, or future generation a burden that you would reject as the condition of your own life.

The fuller place-based argument — including scaling capacity rather than models — is developed in From Climate Intervention to Bioregional Stewardship. Here, neighbouring matters because constitutional principles become real only through the places in which people meet consequences.

14. Rule of Law and Rule of Life

Rule of Law is one of the great achievements of legal civilisation. Power is not supposed to be arbitrary. Authority must be bound by law, procedure, rights, accountability, evidence, and review.

> **No person, office, institution, or authority stands above the law.**

Rule of Law protects persons and communities from arbitrary power.

The planetary crisis reveals a second requirement: a system may be lawful and still life-destroying.

A contract may be valid and still degrade water. A permit may be legal and still destroy habitat. A regulated financial instrument may accelerate extraction. A compliant supply chain may externalise damage into soil, workers, rivers, communities, and future generations. A public decision may follow procedure and still erode the conditions on which law itself depends.

Rule of Life means that law, economy, governance, and technology remain bound to the conditions that make life possible.

> **No economy, institution, technology, or legal form may exclude life-supporting consequences from its account.**

The relation between the two principles is:

> **Rule of Law protects persons from arbitrary power.**
> **Rule of Life protects the living world from lawful destruction.**

Or more poetically:

> **No ruler above the law. No account above life.**

Here an account means both a record of value and a representation of reality. The line is a poetic reminder of the test, not a power to set aside legal rights or declare a single account of life authoritative.

Rule of Life does not reject law. It deepens the constitutional question. Rule of Life without Rule of Law may become arbitrary moralism. Rule of Law without Rule of Life may legitimate lawful extraction.

Planetary governance requires both: lawful process and life-bound consequence.

This is not a simple hierarchy in which “life” mechanically defeats every right or institution. Human life, non-human life, dignity, freedom, livelihood, property, public authority, ecological integrity, and future capacity may conflict. Rule of Life names the requirement that such conflicts cannot be adjudicated as though life-supporting conditions were merely external interests among others.

The constitutional task is to develop lawful institutions capable of hearing ecological consequence without escaping legal constraint.

Rule of Life is therefore better understood as a **constitutional test** than as a superior sovereign rule.

It has two related uses. Within Spiralweb, it is a proposed horizon for the association’s own decisions, ledgers, working methods, and use of AI: its stewards can pause or change their own participation. In relation to decisions held by others, it can offer evidence, questions, and critical friendship without claiming the authority to reverse a lawful decision. Public scrutiny and access to review do not depend on an invitation to friendship. Rule of Law remains a binding constraint in either setting.

It asks whether an otherwise lawful decision is degrading the conditions that make lawful, social, and ecological life possible — and requires those consequences to become visible, contestable, and answerable within legitimate institutions.

It does not supply the answer in advance.

Nor does invoking “life” settle conflicts among rights, livelihoods, ecological integrity, public authority, human need, and future capacity.

Its constitutional function is narrower and more demanding:

> **life-supporting consequence may not be excluded from the account merely because the action producing it is otherwise lawful.**

15. Rule of Life is not ecological exception

Rule of Life is not a licence to suspend Rule of Law in the name of a higher moral order. It is not ecological emergency power, charismatic authority, spiritual vitalism, technocratic optimisation, or a new sovereign exception.

No person or institution may simply declare “life” and thereby override evidence, rights, procedure, proportionality, review, or dissent.

History contains many examples of higher-order language used to release power from constraint: nation, destiny, race, class, security, civilisation, development, God, and emergency. “Life” must not become another word that permits authority to escape accountability.

PG Ledger matters here as an evidentiary and responsibility layer. It can make life-supporting conditions, burdens, uncertainty, decisions, and consequences visible and contestable. It does not decide by itself and does not replace courts, democratic institutions, professional knowledge, or public law.

The test needs more than one kind of reference point: applicable legal and scientific standards; locally agreed conditions for pause or redesign, including steward viability; and procedural safeguards such as stated evidence status, recorded disagreement, and Unknown that stays Unknown. The ledger can hold these side by side, with dates and later returns. It cannot choose the threshold or pronounce a legal outcome. The test is repeated as consequences emerge, rather than exhausted at the moment of initial approval.

Rule of Life becomes legitimate only when bound to:

- documented evidence and claim status;
- lawful procedure and public reasoning;
- rights and proportionality;
- human responsibility;
- contestability and access to correction;
- review of impacts and unintended burdens;
- the possibility of pause, remedy, and reversal.

Legal anchoring

Rule of Life does not begin outside existing law. It can be read in conversation with legal developments already under way: environmental information rights, public participation, access to justice, sustainability disclosure, double materiality, nature restoration duties, human-rights recognition of a clean, healthy, and sustainable environment, and corporate obligations to identify and address ecological and social impacts.

Relevant anchors include the Aarhus Convention; the UN General Assembly's recognition of the right to a clean, healthy, and sustainable environment; the EU Nature Restoration Regulation; CSRD and ESRS; the EU Corporate Sustainability Due Diligence Directive as amended, including Directive (EU) 2026/470 and the consolidated March 2026 text; and the EU AI Act where AI systems affect rights, accountability, and public decision contexts.

These regimes do not amount to a constitutional Rule of Life, and they differ in scope, enforceability, institutional design, and political direction. Law also moves in more than one direction: Directive (EU) 2026/470 narrowed the scope of corporate sustainability reporting and due diligence while other environmental and participation duties remain. The existence of a legal anchor does not by itself make a consequence visible or give those affected the capacity to contest it.

Rule of Law asks: was authority legitimate, procedure followed, evidence handled properly, rights respected, and the decision reviewable?

Rule of Life asks: did the action support or degrade soil, water, bodies, communities, steward viability, ecological integrity, and future capacity — and are those consequences visible enough to contest?

The future constitutional problem is not only how to prevent rulers from becoming arbitrary. It is also how to prevent lawful systems from destroying the living conditions that make lawful society possible.

This may also be read as a constitutional dimension of bottom-linked governance: inquiry begins with those living with the consequences, while rights, procedure, public responsibilities, and review remain with legitimate institutions. AI may help a small practice articulate its own observations in institutionally legible terms; a ledger can preserve the source, uncertainty, burden, and return. Neither creates public authority or relieves public institutions of their duties.

From constitutional test to bounded action

Before a consequential next step, and again when its effects can be observed, the people responsible can ask:

1. **Who and what is affected?** Name the people, species or ecological relations, and places within reach of the action, including connected water, habitat, and future users where relevant.
2. **What is known?** Keep direct observation, specialist assessment, interpretation, disagreement, and Unknown distinguishable. State the scale and limits of each claim.
3. **Who decides and who carries the burden?** Record the responsible human or institution, the people whose lives or work may be changed, and their routes to question, refuse where they have that standing, or seek pause and review.
4. **What can be done and revised?** Choose a step within the field's ecological and human capacity, subject to applicable law and binding limits, with a clear reason to slow, repair, or stop.
5. **When and to whom does the account return?** Agree when consequences will be revisited with those affected, who records what changed, and how the next decision can be corrected.

A species cannot give human consent. Its conditions must instead be approached through situated ecological observation, relevant expertise, applicable protections, and people and institutions accountable for the place. Neither a steward nor a ledger has authority to speak in the species' place. The constitutional task is to keep its living conditions within the account and the resulting human decision open to evidence and challenge.

16. Tragedy, Romance, and Ledger of the Commons

Tragedy

The Tragedy of the Commons warns that shared resources may be depleted where individual incentives reward use while costs are dispersed. The formulation became influential because it named a real danger: commons can collapse.

But tragedy is not destiny. Elinor Ostrom and many communities demonstrated that shared resources can be governed through boundaries, rules, monitoring, graduated responses, conflict resolution, local knowledge, and polycentric institutions.

The relevant lesson is not that commons fail or that privatisation and central command are the only alternatives. It is that commons require governance.

Romance

The opposite error is what this paper calls Romance of the Commons: the belief that community, love of place, tradition, participation, or shared intention will carry the work without sufficient attention to burden, power, money, conflict, and institutional memory. The concern has a clear predecessor in Agrawal and Gibson's study of community and conservation (1999), which challenges idealised accounts of community and directs attention to internal differences, interests, and institutions.

Romance is not trivial. Long stewardship is rarely carried by metrics alone. People care for a river, species, garden, neighbourhood, future, memory, child, or piece of land because something in the field calls them. Devotion is real infrastructure.

But romance without form can become fog.

Fog hides unpaid coordination, gendered labour, informal authority, exhaustion, charismatic capture, land relations, money, data extraction, story ownership, and who is repeatedly asked to wait. A romance of the commons without review can become another form of extraction.

At the same time, ledger without romance becomes dead bureaucracy. A system that only measures, classifies, audits, and reports may prevent some harms but cannot generate devotion. It cannot make a person love the field. It cannot produce the warmth that makes rotation meaningful.

Romance without ledger can become fog.

Ledger without romance can become dead bureaucracy.

Together they can become stewardship.

The commons must be loved, but love must receive form. It must be governed, but governance must remain warm. It must be open enough for belonging and bounded enough to resist capture. It must be documented without being reduced to documentation.

Encounter

Before a commons becomes a ledger entry or a governance decision, people must often be able to meet.

This sounds ordinary.

It is institutional infrastructure.

A hospitable meeting surface is neither a tribunal nor a classroom in which one discipline, institution, funder, technology, or worldview arrives already holding the legitimate vocabulary.

It allows different forms of knowledge, interest, uncertainty, attachment, and disagreement to become present without requiring immediate synthesis.

Hospitality does not mean the absence of boundaries, expertise, evidence, conflict, or authority.

It means that entry into inquiry does not first require epistemic surrender.

A useful table can therefore do something a database cannot: allow people to remain in relation long enough to discover what the disagreement, connection, or shared question actually is.

The commons needs such conditions for encounter alongside devotion, memory, and governance.

Not every difference will become commensurable.

Not every meeting will produce agreement.

The point is more practical:

> **we do not need to solve epistemic difference before creating better conditions for meeting across it.**

The ledger can then preserve what becomes sufficiently clear to carry forward without pretending to contain everything that happened at the table.

Ledger of the Commons

Between Tragedy and Romance stands the Ledger.

The Ledger of the Commons is the memory surface of shared life. It records enough to prevent fog. It distinguishes realities so one gain cannot conceal another collapse. It protects small observations from disappearance without inflating them into universal truth. It makes steward burden visible. It allows money and institutional support to enter without silently becoming sovereign.

A living ledger helps a field say: not yet; too much; too fast; wrong channel; missing consent; disputed evidence; unresolved burden; not for sale; pause and repair.

Regenerative systems can be damaged by too little support, but also by too much support delivered too quickly.

A dry riverbed may need water. If the water arrives as flood, it destroys the channel it intended to revive. The same applies to money, volunteers, media attention, data systems, institutional partnerships, and political ambition.

> **Gold before bloom.** The field must develop enough substrate, rhythm, trust, knowledge, and governance to receive abundance without being overtaken by it.

The problem is not movement. It is acceleration without metabolism. It is scale without absorption. It is capital entering before the field has enough membranes to remain itself.

A ledger-supported commons does not reject flow. Human stewards govern flow through evidence, judgement, consent, boundaries, and correction.

The components now meet:

- Penguin Economics gives the survival logic: rotate warmth toward exposure.
- PG Ledger gives memory and form: connect burdens, flows, evidence, decisions, and corrections.
- Regenerative Reciprocity gives the circulation principle: abundance moves only when life is stable enough to give.
- Rule of Life provides the constitutional horizon: no lawful or profitable arrangement is sufficient if it destroys life-supporting conditions.

17. Federation, failure modes, finance, and structural decency

Federation, not aggregation

The dominant economy scales through fungibility. Money moves rapidly across contexts because it is abstract, exchangeable, and substantially indifferent to place. This supports coordination at scale, but it also allows ecological burden, human exhaustion, and local meaning to disappear behind price.

The stewardship architecture deliberately limits fungibility. Land and ecology, steward viability, and governance capacity are not exchangeable tokens. They cannot be compressed into one success score without losing the information that protects the field.

This creates coordination costs. The system will be slower than extractive capital. It requires translation, review, and situated judgement. It will not outcompete global finance on abstraction, speed, or frictionless scale.

That is not a defect. It is a design commitment.

A Planetary Stewardship Economy scales through federation: many situated ledgers, AnchorPoints, local circles, public institutions, and field-specific membranes connected by shared protocols without being dissolved into one central metric.

Its scaling logic is translation between living contexts, not standardisation without remainder.

Failure modes of stewardship

A credible architecture must turn its own discipline back on itself. Stewardship can also fail.

- **Local gatekeeping:** stewards or circles may reproduce hierarchy, restrict access, or speak for a field without sufficient consent.
- **Ledger overload:** documentation may consume the capacity it was intended to protect.
- **Dashboard reduction:** colours may become rankings or moral scores instead of prompts for judgement.
- **AI legibility bias:** machine-readable signals may displace field-important realities that resist standardisation.
- **Funder capture:** clean reports and scalable claims may be rewarded over difficult truth.
- **Pause misuse:** Red Phase or stop authority may be used to delay legitimate criticism or protect insiders.
- **Decency as pressure:** anstændighed may be moralised into conformity if not bound to evidence, process, rights, and conflict resolution.
- **Federation conflict:** connected fields may develop incompatible claims, interests, and interpretations that goodwill alone cannot resolve.
- **Commons romance:** love of place may hide unpaid labour, emotional obligation, gendered burden, charismatic authority, or informal extraction.

These failure modes do not invalidate the architecture. They define its next level of governance. The ledger must be able to record the strains and correction needs of the stewardship system itself.

Financial claims and living-system capacity

Credit can finance useful work long before its benefits arrive. It can also place demands on future income that a household, business, or public institution cannot sustain. The question is what happens when meeting those demands takes priority over the conditions that make income and shared life possible.

Minsky gives a precise account of one financial danger. In his Financial Instability Hypothesis (1992, pp. 6–8), Ponzi finance describes units whose operating cash flows cannot cover either principal or interest payments, so that they must borrow further or sell assets. This is a financing position, not a finding of fraud and not a description of every part of the economy. It identifies how dependence on refinancing can make a system fragile.

Money creation does not remove that difficulty. As the Bank of England explains, commercial-bank lending creates deposits; central-bank interest-rate policy and asset purchases operate through different mechanisms. More money or credit does not itself restore depleted soil, produce rested people, or decide who should absorb a loss.

The stewardship question concerns that distribution of consequences. When creditors, owners, or institutions defend expected returns, who carries the adjustment: investors, taxpayers, workers, public services, or the living field? Which arrangements permit losses to be shifted onto people and places with the least power to refuse? These questions require an account of the particular debt, decision, and burden; a general accusation against banking cannot answer them.

> **Financial claims become extractive when their enforcement consumes the ecological and human capacity on which repayment and shared life depend. A stewardship economy asks which claims must change so that the living field can continue.**

Penguin Economics proposes structural inversion:

- where the dominant system accelerates intake, ask about absorption;
- where it rewards scale, ask about field viability;
- where it externalises burden, bring burden into view;
- where it converts life into collateral, protect the field from ownership capture;

- where it asks how value can be extracted, ask whether the conditions of life remain intact.

Steward viability across a life course

Steward viability cannot end at the boundary of active service. A mature stewardship economy must eventually address sickness, rest, ageing, insurance, succession, pension-like continuity, dignified exit, and intergenerational transfer of field knowledge.

This is future institutional work rather than a fully developed model in this paper. It must nevertheless be named. A system that protects soil while abandoning the ageing steward has not completed the logic of Rule of Life.

Anstændighed as structural decency

A central Danish word in this architecture is anstændighed. “Decency” is close but often too weak. “Integrity” is close but too formal. In this context, anstændighed means conduct and institutional form that do not destroy the conditions for shared survival.

It is the minimum structural decency required for shared life to continue.

> Do not take warmth while assigning cold to others.
> Do not call a field successful while hiding who carries the burden.
> Do not make ecological ambition depend on unpaid exhaustion.
> Do not use commons language to cover extraction.
> Do not treat legal permission as moral sufficiency.
> Do not demand reciprocity before viability.
> Do not confuse speed with life, visibility with truth, money with capacity, or openness with justice.

In Penguin Economics, anstændighed is not an optional virtue. It is a survival condition.

Closing diagnostic

The final test is simple and severe:

- Does this action help life continue?
- Does it protect the field from capture?
- Does it make burden visible?
- Does it rotate warmth toward exposure?

- Does it make situated observation legible without replacing human judgement?
- Does it allow money to enter without becoming sovereign?
- Does it keep carbon, biodiversity, and financial metrics inside a wider account rather than making one metric sovereign?
- Does it respect Earth Time rather than forcing living systems into investor, electoral, project, platform, credit, or reporting time?
- Does it protect the carrying capacity of bodies, relationships, institutions, and civic systems?
- Can the people carrying the work describe its costs in their own words, contest its account, and sustain a decent livelihood?
- Does it allow abundance to circulate only when life is strong enough to give?
- Can the action be questioned, corrected, paused, and reviewed under named human responsibility?

If yes, it may belong to a Planetary Stewardship Economy.

If no, it may still be legal, profitable, philanthropic, innovative, compliant, or well-branded. But it is not Penguin Economics.

For now, these are questions for a living inquiry. One founding steward's experience can expose a real burden and help develop a workable method; it cannot establish that other people will benefit in the same way. The hypothesis must be revised if AI narrows whose experience counts, if institutional legibility transfers authority away from a place, or if the centre grows by consuming the time and livelihood of those who hold it.

Closing

The old constitutional question was how to govern property, contract, sovereign power, and the relation between authority and person.

The emerging question is how to govern the conditions of life without turning "life" into an excuse for arbitrary power.

Rule of Law remains necessary because arbitrary power destroys trust, freedom, and justice.

Rule of Life becomes necessary because lawful extraction can destroy the world on which law depends.

The commons must not be abandoned to tragedy. Nor may it dissolve into romance. It must be loved enough to be governed and governed warmly enough to remain alive.

That is the work of the Ledger of the Commons.

That is the survival intelligence of Penguin Economics.

That is the constitutional horizon of Rule of Life.

That is the beginning of a Planetary Stewardship Economy.

References

Agrawal, A., & Gibson, C. C. (1999). Enchantment and Disenchantment: The Role of Community in Natural Resource Conservation. World Development, 27(4), 629–649. doi: 10.1016/S0305-750X(98)00161-2.

McLeay, M., Radia, A., & Thomas, R. (2014). Money creation in the modern economy. Bank of England Quarterly Bulletin, 2014 Q1.

Minsky, H. P. (1992). The Financial Instability Hypothesis. Working Paper No. 74. Levy Economics Institute.

Trust-Based Philanthropy Project. Six grantmaking practices. Consulted 17 October 2026.

This paper is a conceptual and operational synthesis. The references below provide an anchor layer rather than a comprehensive literature review.

Bonney, R., Cooper, C. B., Dickinson, J., Kelling, S., Phillips, T., Rosenberg, K. V., & Shirk, J. (2009). Citizen science: A developing tool for expanding science knowledge and scientific literacy. BioScience, 59(11), 977–984.

Convention on Biological Diversity. (2024). Revised guidance on using the indicators of the monitoring framework of the Kunming–Montreal Global Biodiversity Framework (CBD/COP/16/INF/3/Rev.1).

Daly, H. E. (1996). Beyond Growth: The Economics of Sustainable Development. Beacon Press.

European Commission. (2023). Commission Delegated Regulation (EU) 2023/2772 supplementing Directive 2013/34/EU as regards sustainability reporting standards.

European Parliament and Council. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act).

European Parliament and Council. (2024). Regulation (EU) 2024/1991 on nature restoration.

European Parliament and Council. (2024/2026). Directive (EU) 2024/1760 on corporate sustainability due diligence, consolidated version of 18 March 2026.

European Parliament and Council. (2026). Directive (EU) 2026/470 amending corporate sustainability reporting and due diligence requirements.

Federici, S. (2012). Revolution at Point Zero: Housework, Reproduction, and Feminist Struggle. PM Press.

Folbre, N. (2001). The Invisible Heart: Economics and Family Values. The New Press.

Foundational Economy Collective. (2018). Foundational Economy: The Infrastructure of Everyday Life. Manchester University Press.

García, M., Eizaguirre, S., & Pradel, M. (2015). Social innovation and creativity in cities: A socially inclusive governance approach in two peripheral spaces of Barcelona. City, Culture and Society, 6(4), 93–100.

Hardin, G. (1968). The tragedy of the commons. Science, 162(3859), 1243–1248.

Hickel, J. (2020). Less Is More. William Heinemann.

IPBES. (2019). Global Assessment Report on Biodiversity and Ecosystem Services.

Irwin, A. (1995). Citizen Science: A Study of People, Expertise and Sustainable Development. Routledge.

Sakschewski, B., et al. (2025). Planetary Health Check 2025: A Scientific Assessment of the State of the Planet. Potsdam Institute for Climate Impact Research.

Mang, P., & Reed, B. (2012). Designing from place: A regenerative framework and methodology. Building Research & Information, 40(1), 23–38.

Meadows, D. H. (2008). Thinking in Systems: A Primer. Chelsea Green.

Moulaert, F., MacCallum, D., Van den Broeck, P., & García, M. (2019). Bottom-linked governance and socially innovative political transformation. In Atlas of Social Innovation: 2nd Volume — A World of New Practices, 62–65.

Ostrom, E. (1990). Governing the Commons. Cambridge University Press.

Ostrom, E. (2010). Beyond markets and states: Polycentric governance of complex economic systems. American Economic Review, 100(3), 641–672.

Raworth, K. (2017). Doughnut Economics. Chelsea Green.

Richardson, K., et al. (2023). Earth beyond six of nine planetary boundaries. Science Advances, 9(37), eadh2458.

Rockström, J., et al. (2009). A safe operating space for humanity. Nature, 461, 472–475.

UNECE. (1998). Convention on Access to Information, Public Participation in Decision-making and Access to Justice in Environmental Matters (Aarhus Convention).

United Nations General Assembly. (2022). The human right to a clean, healthy and sustainable environment (A/RES/76/300).

Spiralweb / Green Papers architecture

Engberg, L. A. (2026). Moral Biology. Green Paper 01.

Engberg, L. A. (2026). Civic Nervous System.

Engberg, L. A. (2026). Land, Water and Living Stewardship.

Engberg, L. A. (2026). Authority, Pause, Refusal and Correction.

Engberg, L. A. (2026). Silent Degradation.

Engberg, L. A. (2026). Sophia Lumen: A relational human–AI working practice.

Engberg, L. A. (2026). Sophia Lumen Protocol: A revisable protocol for human–AI co-inquiry. Candidate protocol v0.1.

Engberg, L. A. (2026). The Correction Loop: Preserving Human Accountability in AI-Assisted Work. Report 02.

Engberg, L. A. (2026). Water Into Dry Riverbeds. Report 03.

Engberg, L. A. (2026). Penguin Dashboard: Legibility as Governance. Report 04.

Engberg, L. A. (2026). SRIP: The Steward’s Journey. Report 05.

Engberg, L. A. (2026). Regenerative Reciprocity. Report 06.

Engberg, L. A. (2026). Knowing From the Ground.

Engberg, L. A. (2026). From Climate Intervention to Bioregional Stewardship.

Green Papers / Spiralweb · CC BY 4.0